# Scikit-learn Linear Models – Complete Crisp Guide

This is a **complete, structured, crisp explanation of every section** you pasted. Think of this as a **mental model** + **interview** + **production cheat-sheet**. If you truly understand this page, you are *very strong* in classical ML.

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## 1. What are Linear Models (Big Picture)

**Idea**: Predict output as a *linear combination of features*.

$$\hat{y} = w_0 + w_1 x_1 + \cdots + w_p x_p$$

- `coef_` → weights $w_1 \ldots w_p$
- `intercept_` → bias $w_0$

✅ Used for:

- Regression (predict numbers)
- Classification (via link functions like sigmoid)

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## 2. Ordinary Least Squares (OLS)

### What it does

Minimizes **sum of squared errors**:

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$$\min_w ||Xw - y||_2^2$$

### Properties

✅ Fast, closed-form solution (via SVD) ❌ Sensitive to **multicollinearity** (correlated features) ❌ High variance when X is nearly singular

### When to use

- Small to medium datasets
- Low feature correlation
- Baseline model

### Complexity

$O(n_{samples} \cdot n_{features}^2)$

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## 3. Non-Negative Least Squares (NNLS)

### Idea

Force all coefficients ≥ 0

### Why useful

- Physical quantities
- Counts, prices, intensities

### How

Python

```python
LinearRegression(positive=True)

```

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## 4. Ridge Regression (L2 Regularization)

### Problem it solves

OLS explodes when features are correlated.

### Objective

$$||Xw - y||_2^2 + \alpha||w||_2^2$$

### Effect

- Shrinks coefficients
- Keeps all features
- Improves numerical stability

### Alpha intuition

- α = 0 → OLS
- Large α → smaller weights

### Regression vs Classification

- `Ridge` → regression
- `RidgeClassifier` → classification via sign of prediction

✅ Faster than LogisticRegression for many classes

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## 5. RidgeCV / RidgeClassifierCV

### Purpose

Automatically select **best alpha**

### Default

- Leave-One-Out Cross-Validation (LOO)

⚠️ alpha cannot be 0 with LOO

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## 6. Lasso (L1 Regularization)

### Objective

$$\frac{1}{2n}||Xw - y||^2 + \alpha||w||_1$$

### Key Property

✅ Produces **exact zeros** → Feature Selection

### Algorithm

- Coordinate Descent
- Soft-thresholding

### When to use

- You want sparse model
- Feature selection matters

❌ Unstable with correlated features

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## 7. Coordinate Descent + Gap Safe Screening

### Coordinate Descent

- Update **one coefficient at a time**
- Efficient for L1 problems

### Gap Safe Screening

- Detect features guaranteed to be zero
- Remove them early → speed

✅ Makes Lasso scalable

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## 8. Choosing Alpha for Lasso

### Options

1. `LassoCV` → Cross-validation (preferred)
2. `LassoLarsCV` → Small sample, many features
3. `LassoLarsIC` → AIC / BIC (cheap but risky)

# 9. AIC / BIC (Model Selection Theory)

## Idea

Balance:

- Fit quality
- Model complexity
- AIC → lighter penalty
- BIC → stronger penalty

⚠️ Works best when:

- n_samples > n_features
- Model well-conditioned

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# 10. Elastic Net (L1 + L2)

## Objective

$$L = L_{OLS} + \alpha(\rho||w||_1 + (1-\rho)||w||_2^2)$$

## Why it exists

- Lasso drops correlated features randomly
- Ridge keeps all
- ElasticNet keeps **groups** of correlated features

✅ Best default for real data

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# 11. Multi-Task Lasso / ElasticNet

## Scenario

Multiple related targets (tasks)

### Constraint

Same features selected across tasks

✅ Time series, multi-output problems

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## 12. Least Angle Regression (LARS)

### Idea

- Like forward selection
- Adds features gradually
- Produces full regularization path

✅ Efficient when features >> samples

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## 13. LassoLars

- Exact Lasso solution
- Piecewise linear path
- Great for analysis

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## 14. Orthogonal Matching Pursuit (OMP)

### Constraint

Limit number of non-zero coefficients

### Nature

- Greedy
- Approximate

✅ Sparse signal recovery

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## 15. Bayesian Regression (Big Picture)

### Philosophy

Treat weights as **random variables**

✅ Automatic regularization ✅ Uncertainty estimation

❌ Slower

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# 16. Bayesian Ridge Regression

- Ridge with probabilistic interpretation
- Estimates α and λ from data
- Robust to ill-posed problems

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# 17. ARD Regression (Sparse Bayesian Learning)

## Difference from Bayesian Ridge

- One variance per feature
- Automatically prunes irrelevant features

✅ Bayesian alternative to Lasso

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# 18. Logistic Regression

## Important truth

Logistic Regression = **Linear Classifier**

## Output

Probability via sigmoid:

$$p = \frac{1}{1 + e^{-(Xw + b)}}$$

## Regularization

- L1, L2, ElasticNet
- Controlled via C (= inverse regularization)

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# 19. Logistic Regression Solvers

## Key intuition

- `lbfgs` → safest default
- `liblinear` → small, binary
- `saga` → large, sparse, ElasticNet
- `newton-cholesky` → n_samples >> n_features

✅ Solver choice = performance

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# 20. Generalized Linear Models (GLM)

## Extension

Linear predictor + link function + non-Gaussian loss

## Examples

- Poisson → counts
- Gamma → positive skewed
- Bernoulli → probabilities

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# 21. Tweedie Regressor

Single model for:

- Normal
- Poisson
- Gamma
- Inverse Gaussian

✅ Industry favorite (insurance, risk)

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# 22. SGD, Perceptron, Passive-Aggressive

### Use case

- Huge data
- Online learning
- SGD → flexible
- Perceptron → simplest
- PA → aggressive updates

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# 23. Robust Regression

## Goal

Handle outliers

### Models

- RANSAC → large y-outliers
- Theil-Sen → X-outliers
- Huber → smooth transition

✅ When data is dirty

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# 24. Quantile Regression

## Difference from OLS

Predicts **median / quantiles**, not mean

✅ Prediction intervals ✅ Heteroscedastic noise

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# 25. Polynomial Regression

## Trick

Make data nonlinear, model stays linear

✅ Feature engineering technique ✅ Use with pipelines

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# FINAL MENTAL MAP (Remember This)

| | Problem | Best Model |
|---|---|---|
| 1 | Clean regression | OLS |
| 2 | Multicollinearity | Ridge |
| 3 | Feature selection | Lasso |
| 4 | Correlated features | ElasticNet |
| 5 | Probabilistic | Bayesian Ridge |
| 6 | Sparse Bayesian | ARD |
| 7 | Classification | Logistic |
| 8 | Counts | Poisson |
| 9 | Outliers | Huber / RANSAC |
| 10 | Huge data | SGD |

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If you want next: ✅ **Interview Q&A** ✅ **Real-world decision tree** ✅ **Hands-on notebook roadmap** ✅ **Production pitfalls**

Tell me how deep you want to go 👇